

Advancing Legal Reasoning in Algerian Law

RAG + Knowledge Graphs + Argument Mining → Citation-Faithful legal answers

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WHY
01

The Problem

WHY LEGAL QA IS HARD

1.1 Fluent is not lawful

A legal answer cannot be judged by fluency or plausibility. To be valid it must be **grounded in identifiable sources**, cite the **exact governing articles**, stay inside the **Algerian jurisdiction**, and read each provision **as amended** at the relevant date — and where the law is silent, the only correct answer is a **refusal**. A fluent language model guarantees none of this.



SOURCED · ARTICLE-CITED · JURISDICTION-BOUND ·
AMENDMENT-AWARE · OR REFUSED

1.2 What legal questions demand

بَيِّن الأساس القانوني لمبدأ خضوع القاضي للنص
التشريعي، وحدود سلطته عند غياب النص...

— demands **exact articles** — Civil Code, arts. 1–2

كيف تغيّرت شروط الخلع في قانون الأسرة بعد تعديل
؟2005

— demands the law **as amended** — Ord. 05-02 → Family
Code, art. 54

هل يحق لصاحب العمل في الجزائر فصل العامل بحرية
تامة دون إبداء أسباب (at-will)؟

— a **foreign-doctrine trap** — Algerian law is silent; the system
must **abstain**

verbatim from AlgerianLegalBench (244 questions)

1.3 Where today's LLMs fail

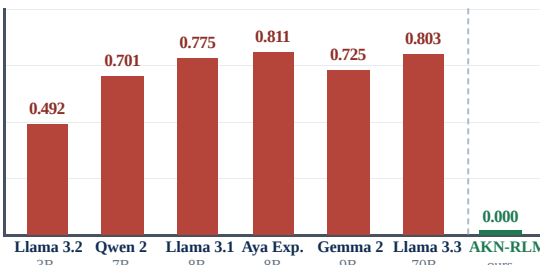


Fig. 1. Hallucinated-citation rate: the share of cited articles that *do not* exist, per direct-LLM baseline (244 questions). Scale does not help — the 70B model fabricates 4 of 5 citations. Gemini 2.5 reaches 0.008 only by declining to cite at all, and six of seven baselines import foreign law on >12 of 40 trap questions. **AKN-RLM: 0.000.**

1.4 ...and Algerian law raises the bar

- **Arabic-first.** 232 of 244 benchmark questions are in Arabic
- **Layered in time.** 1963–2025; amendments rewrite articles in place
- **Deeply structured.** code → chapter → section → article
- **Low-resource.** barely covered by existing legal-AI corpora

1.5 Objectives

- ✓ **Structure the law**
171 docs → Akoma Ntoso XML + legal KG
- ✓ **Benchmark it**
AlgerianLegalBench: 244 expert questions
- ✓ **Reason on evidence**
typed retrieval + bounded recursion
- ✓ **Verify or abstain**
triple faithfulness gate

1.6 Significance & impact

- First citation-faithful QA system for Algerian law** — every claim resolves to a registered article, or the system abstains.
- Reusable infrastructure for Algerian legal AI**
 - **AKN corpus + knowledge graph** — 171 structured documents, 1963–2025, article-level registry
 - **AlgerianLegalBench** — 244 expert-annotated questions, $\kappa = 0.829$
- A recipe for low-resource legal AI** — evidence control beats parameter scale.

HOW
02

The AKN-RLM Architecture

DOCUMENTS → STRUCTURE → VERIFIED ANSWER

Four zones, each repairing a failure mode of the last — a controlled evidence pipeline, not one model.

1 STRUCTURE THE LAW

171 files · 1963–2025
FRBR article registry



Raw texts
PDF · TXT



Akoma Ntoso
XML · articles



Knowledge Graph
RDF · amend. chains

2 RETRIEVE EVIDENCE

RRF(BM25 + Dense)



Legal question
AR · FR



Classifier
routes 8 types

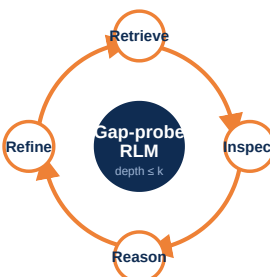


Hybrid retrieval
typed handlers

exact-article rule-application multi-hop temporal conceptual layman
long-context unanswerable

3 REASON RECURSIVELY

bounded · gap-driven
depth-limited



- **Detects evidence gaps**
insufficient support → expand
- **Toulmin ADU mining**
on every cited article
- **One corrective retry**
then stops — no open recursion

✓ Evidence sufficient → verify & answer

4 VERIFY & ANSWER

triple faithfulness gate



Citation exists
vs registry



Jurisdiction OK
canary check



Claim supported
multiling. NLI



Verifiable answer
every claim → a registered article



Principled abstention
when the law is silent

PROOF
03

Results

FROZEN 244-QUESTION RUN

0.305

Citation
F1

locked
deployable

✓ 0.000 HCR
· JIR

0.703

✓ Abstention
F1

1.64×

vs direct LLM

2.9×

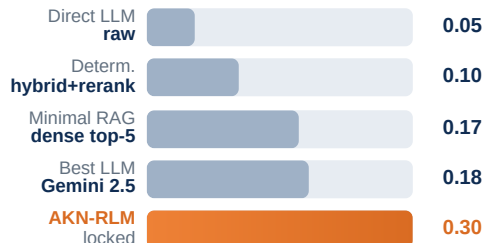
vs determ. retr.

1.74×

vs minimal RAG

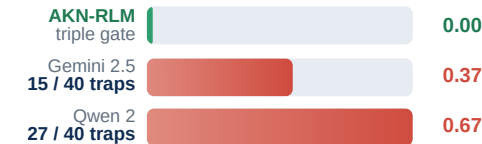
Citation F1 vs baselines

HIGHER = BETTER



Jurisdictional infection (JIR)

LOWER = SAFER



Only AKN-RLM holds **0.000** hallucinated-citation & foreign-law rates. Reliability tracks **legal structure, not scale** — the largest baselines hallucinate most.

AlgerianLegalBench

244 QUESTIONS

23

CATEGORIES

8

QUERY TYPES

40

INFECTION TRAPS

232

ARABIC Q

12

FRENCH Q

.829

K AGREEMENT

Difficulty mix

Easy · Medium · Hard



LIMITS

- Retrieval ceiling: R@10 = 0.258
- French under-served
- Abstention precision 0.654
- Metrics are proxies

NEXT

- Corpus-tuned embedder
- French ↔ Arabic retrieval
- Graph-retrieval redesign
- Expert legal review



Structured sources

+



Controlled retrieval

+



Recursive reasoning

+



Citation verification

=

THE TAKEAWAY

Trustworthy Legal AI

an **evidence-controlled system**, not an unconstrained generator.